

EART11200 - The Natural Scientist's Toolkit
Mathematics and Physics

Week 1 Practical - Calculating Gradients (Solutions)

Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

(1) Discuss the role of calculus in broader mathematical and physical contexts.

(2) Apply Euler's Backward Method to analyse rates of change in mathematical functions and real world applications.

(3) Apply Euler's Forward Method to analyse rates of change in mathematical functions and real world applications.

(4) Apply Central Differences Method to analyse rates of change in mathematical functions and real world applications.

Exercise 1: Rearranging expressions

- In many cases we will be required to rearrange formulae.
- Rearranging formulae is a way of changing the subject of a formula.
- This can help us determine a missing value when we know other values within a formula.
- To change the subject of a formula is to rewrite the formula so a single unknown variable is equal to the rest of it.
- For example, if $y = \sqrt{x+2}$ then we can rearrange to make x the subject (i.e. $x = y^2 - 2$)

Exercise 1: Rearranging expressions

- The equation describing radioactive decay is:

$$\frac{dN}{dt} = -\lambda N$$

where N is the number of unstable nuclei, t is time and λ is called the decay rate.

- This is a differential equation, and it has solution:

$$N(t) = N_0 \exp(-\lambda t)$$

where N_0 is the total number of nuclei in the sample at $t = 0$.

- The half-life is the time it takes for the number of unstable nuclei to decrease by one half (i.e. at $t_{1/2}$, then $N(t_{1/2}) = N_0/2$.
- Show that the decay rate is

$$\lambda = \frac{\ln(2)}{t_{1/2}}$$

Solution 1: Rearranging expressions

- At the half-life, $t = t_{1/2}$, the number of nuclei is halved so:

$$N(t_{1/2}) = \frac{N_0}{2} \quad \text{and} \quad N(t_{1/2}) = N_0 \exp(-\lambda t_{1/2})$$

- Equating and rearranging gives

$$\begin{aligned} \frac{N_0}{2} &= N_0 \exp(-\lambda t_{1/2}) \\ \frac{1}{2} &= \exp(-\lambda t_{1/2}) \\ -\ln(2) &= -\lambda t_{1/2} \\ \lambda &= \frac{\ln(2)}{t_{1/2}} \end{aligned}$$

Exercise 2: Calculating Gradients

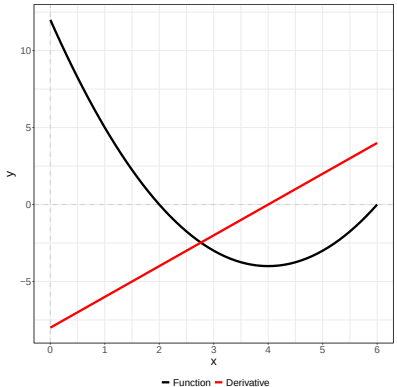
- The plot shows the following function between $x = 0$ and $x = 6$:

$$y = x^2 - 8x + 12$$

and the rate of change (i.e. the derivative):

$$\frac{dy}{dx} = 2x - 8$$

- We want to compare the analytical and numerical derivative.



Exercise 2: Calculating Gradients

- We want to use the **approximation methods** discussed in lectures:
- **Euler's Backward Method**:

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} \equiv \lim_{\Delta x \to 0} \frac{y_x - y_{x-\Delta x}}{\Delta x}$$

- **Euler's Forward Method**:

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} \equiv \lim_{\Delta x \to 0} \frac{y_{x+\Delta x} - y_x}{\Delta x}$$

- **Central Differences Method**:

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} \equiv \lim_{\Delta x \to 0} \frac{y_{x+\Delta x} - y_{x-\Delta x}}{2\Delta x}$$

Exercise 2: Calculating Gradients

(a) Fill in the table below with the approximations to the derivative using an increment of 1.

(b) Fill in the analytical derivative

(c) Do they produce the same answer?

x	y	Backward	Forward	Central	Analytical
0	12				
1	5				
2	0				
3	-3				
4	-4				
5	-3				
6	0				

Function: $y = x^2 - 8x + 12$

Derivative: $\frac{dy}{dx} = 2x - 8$

Backward: $\frac{\Delta y}{\Delta x} = \frac{y_x - y_{x-\Delta x}}{\Delta x}$

Forward: $\frac{\Delta y}{\Delta x} = \frac{y_{x+\Delta x} - y_x}{\Delta x}$

Central: $\frac{\Delta y}{\Delta x} = \frac{y_{x+\Delta x} - y_{x-\Delta x}}{2\Delta x}$

Solution 2: Calculating Gradients

- Filling in the table using the three numerical an analytical methods yields somewhat different results.
- Some methods will be more accurate than others.
- Central differences works better here since it is a more accurate method for smooth functions.

x	y	Backward	Forward	Central	Analytical
0	12				
1	5	-7	-5	-6	-6
2	0	-5	-3	-4	-4
3	-3	-3	-1	-2	-2
4	-4	-1	1	0	0
5	-3	1	3	2	2
6	0				

Function: $y = x^2 - 8x + 12$

Derivative: $\frac{dy}{dx} = 2x - 8$

Backward: $\frac{\Delta y}{\Delta x} = \frac{y_x - y_{x-\Delta x}}{\Delta x}$

Forward: $\frac{\Delta y}{\Delta x} = \frac{y_{x+\Delta x} - y_x}{\Delta x}$

Central: $\frac{\Delta y}{\Delta x} = \frac{y_{x+\Delta x} - y_{x-\Delta x}}{2\Delta x}$

Solution 2: Calculating Gradients

- As we know the analytical derivative, we can compare against the numerical.

$$\% \text{ Error: } 100 \times \left(\frac{\text{Estimate} - \text{Actual}}{\text{Actual}} \right)$$

- At a step size of 1 using the backward method yields a pretty large error.
- As we decrease the step size we get a lower error and closer to the analytical derivative and therefore we see that:

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

Step size $\Delta x = 1$

x	y	Backward	Analytical	Error
0	12			
1	5	-7	-6	14.29%
2	0			

Step size $\Delta x = 0.1$

x	y	Backward	Analytical	Error
0.9	5.61			
1.0	5.00	-6.1	-6.0	1.64%
1.1	4.41			

Step size $\Delta x = 0.01$

x	y	Backward	Analytical	Error
0.99	5.0601			
1.00	5.0000	-6.01	-6.00	0.17%
1.01	4.9401			

Step size $\Delta x = 0.001$

x	y	Backward	Analytical	Error
0.999	5.006001			
1.000	5.000000	-6.001	-6.000	0.02%
1.001	4.994001			

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